

Gazelle Taherzadeh, Ph.D.

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PROFESSIONAL OVERVIEW

Lead Data Scientist with 15+ years of experience designing, developing, and deploying machine learning (ML) systems, predictive models, and computational solutions within a PBM environment supporting ~5 million members across healthcare. Ph.D. in Computer Science focused on computational biology and drug discovery, recognized for developing the first ML-based protein-peptide interaction prediction tool deployed as a publicly available platform, contributing to a CASP/CAPRI docking team ranked among the top 10 internationally, founding a Data Science program from scratch, and receiving the Outstanding New Faculty Award. Currently serving as Lead Applied Data Scientist (Automation manager), leading and building hands-on production AI and data science solutions, including an end-to-end medication adherence management platform integrating predictive modeling, explainability, causal inference, and LLM-based intervention; claims and member KPI trend and forecasting pipelines in reporting dashboards; and enterprise reporting automation achieving 95% report accuracy across 600+ reports while reducing manual reporting effort by 2,000+ hours annually.

U.S. Citizen.

TECHNICAL SKILLS

- **AI, Machine Learning, & Deep Learning:** Structured and unstructured Data Analysis, Machine Learning (ML), Predictive Modeling, Classification, Regression, Clustering, Feature Engineering, Ensemble Methods, Statistical Modeling, Explainable AI (XAI), Scikit-learn (Sklearn), Deep Learning (DL), Graph Neural Networks (GNNs), Convolutional Neural Networks (CNNs), Long Short-Term Memory (LSTM), Natural Language Processing (NLP), Large Language Models (LLMs), Foundation Models, Foundation-Model Pretraining & Fine-Tuning, Large-Model Training at Scale, Multi-GPU / Distributed Training, Retrieval-Augmented Generation (RAG), Prompt Engineering, Model Evaluation, Hypothesis Testing, Survival Analysis, Evolutionary & Genetic Algorithms, TensorFlow, PyTorch, GPU Computing, Keras, LangChain.
- **Computational Biology & Healthcare Analytics:** Proteomics, Structural Biology, Protein Sequence & Structure Analysis, Protein Interaction (PPI) and Function Prediction, Small Biomolecular Analysis and Interaction, Computational Drug Discovery & Design, BLAST, PSI-BLAST, Multiple Sequence Alignment (MSA), PSSM, Rosetta, ZDOCK, PyMOL, Healthcare Analytics, Pharmacy

Informatics, Pharmacy Claims & Benefits, Eligibility and Membership, Medication Adherence, Clinical Data Analysis, Social Determinants of Health (SDoH).

- **Programming, Data & MLOps Engineering:** Python, SQL, Bash(fluent), PySpark, R, MATLAB (working knowledge), NumPy, SciPy, Pandas, Biopython, Distributed Computing, Snowflake, Databricks, Apache Spark, Delta Lake, Unity Catalog, Azure, MLflow, Docker, Git, CI/CD, ETL/ELT Pipeline, Data Pipeline Design, Cloud-Based Data Lake Architecture, Enterprise Data Platform Implementation, Model Deployment & Monitoring.
 - **Data Visualization & BI:** Matplotlib, Seaborn, Power BI, Tableau.
 - **AI Strategy & Architecture:** AI Strategy, AI Solution Architecture, End-to-End AI Development, Responsible AI, Technology & Vendor Evaluation, Roadmap Development.
 - **Technical Leadership:** Technical Team Leadership, Team Management & Mentoring, Cross-Functional Leadership, Executive & Stakeholder Communication.
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PROFESSIONAL EXPERIENCE

PerformRx | Philadelphia, PA

Pharmacy Informatics department within PerformRx, a mid-size PBM subsidiary of AmeriHealth Caritas with ~350 employees, managing prescription drug benefits for Medicaid, Medicare, and Commercial plan members across multiple states.

Lead Applied Data Scientist (Automation manager) | January 2024 - Present

- Architect and lead an end-to-end AI-driven medication adherence management platform, with primary hands-on ownership of development, integrating predictive modeling, explainability, causal analysis, LLMs, RAG, and prompt engineering to support clinical and operational decision-making across ~2 million members.
- Formulate complex healthcare and pharmacy problems as computational questions, characterize heterogeneous data, and design predictive approaches for healthcare outcomes and operational KPI forecasting, including ensemble methods, survival analysis, and feature engineering.
- Implement the end-to-end machine learning lifecycle, including model development, evaluation, registration, deployment, and monitoring through Databricks, Unity Catalog, and Azure.
- Lead a team of developers in automating 600+ enterprise reports across seven business functions, achieving 95% report accuracy, with 50+ reports deployed to production through a collaborative CI/CD workflow.
- Serve as the primary Pharmacy Informatics point of contact for the transition from local SQL Server to a cloud-based Data Lake and Databricks architecture, translating departmental requirements to platform and production teams and tailoring the environment to support AI, analytics, and automation workflows.
- Evaluate emerging AI technologies, including LLMs and RAG, to support build-vs-buy decisions and vendor selection, assessing feasibility, limitations, and applicability to healthcare use cases.
- Translate complex business requirements, analytical findings, model behavior, and uncertainty into actionable recommendations for business stakeholders, directors, and senior leadership.
- Lead a team of five direct reports and collaborate with cross-functional developers and managers across AI/ML, data engineering, reporting, and software development.

Core Stack: MLflow, LangChain, Survival Analysis, Python, SQL, PySpark, Databricks, Azure, Scikit-learn, PyTorch, Power BI, Tableau.

Senior Pharmacy Analyst | August 2023 - January 2024

- Built a prototype for a centralized predictive, interactive, and static reporting tool, laying the foundation for the enterprise-wide report automation initiative.
- Translated business requirements from stakeholders into reporting specifications, analytical frameworks, and report designs across healthcare and pharmacy operations.
- Conducted a systematic assessment of healthcare and pharmacy data sources, data types, frameworks, and reporting structures, developing domain knowledge that directly informed and accelerated the subsequent report automation program.
- Promoted to Automation manager within four (4) months based on demonstrated technical contribution and domain understanding.

Core Stack: Python, SQL, Power BI, Tableau.

Wilkes University | Wilkes-Barre, PA

Private university with ~5,500 students; led applied AI and computational biology research, founded the Data Science program, and bridged academic and industry AI applications.

Visiting Professor | August 2024 - May 2026

- Partnered with four industry organizations to translate real-world AI and software challenges into project scopes, oversaw student delivery, and bridged academic and industry applications.

Assistant Professor & Director of Data Science Program | August 2021 - December 2023

- Founded and built the Data Science program from the ground up, designing curriculum, defining the academic roadmap, and aligning program offerings with industry needs in AI, machine learning, and cloud computing.
- Taught core computer science, AI, machine learning, and data science to 500+ students across undergraduate and graduate programs; received the Outstanding New Faculty Award (2023).
- Applied natural language processing and machine learning to Twitter (X) data, working with large-scale unstructured text for research and analysis.
- Supervised a summer computational biology research program that developed a new protein and small molecular complex dataset, subsequently used in doctoral research developing deep learning approaches for related computational biology projects and contributing to multiple publications in high-impact peer-reviewed journals.
- Collaborated across academic institutions through a multi-institution NSF LSAMP collaboration supporting underrepresented STEM students.

Core Stack: Classification, Forecasting, NLP, Unstructured Data Analysis, Python, TensorFlow, PyTorch, Biopython, Scikit-learn.

University of Maryland, Institute for Bioscience and Biotechnology Research | Rockville, MD

Joint research institute between the University of Maryland and NIST with ~290 members, focusing on biotechnology and computational approaches to biological and therapeutic research, including protein engineering and drug discovery.

Postdoctoral Research Scholar | March 2020 - June 2021

- Developed and applied GNN-, CNN-, and LSTM-based machine learning models for computational drug discovery, including antibody-antigen and T-cell receptor (TCR) molecular interaction modeling and docking, evaluated on a test set of 67 complexes.
- Contributed to research on antibody and T-cell receptor recognition, including SARS-CoV-2 antibody recognition, using large-scale biological datasets, with findings published in *PLOS Computational Biology* (19 citations).
- Contributed to a CASP/CAPRI protein-protein docking team that ranked among the top 10 internationally in a blind prediction competition, achieving performance on a rigorous, independently evaluated scientific benchmark.

Core Stack: Deep Learning, CNN, GNN, LSTM, Python, PyTorch, TensorFlow, Biopython, Bash, Rosetta, ZDOCK, PyMOL.

Griffith University | Gold Coast, Australia

Top-ranked global research university consistently ranked among the top 2% of universities worldwide, with doctoral research focused on machine learning, bioinformatics, and computational molecular science.

Data Scientist | December 2019 - March 2020

- Worked on data analysis and computational implementations for machine learning and biological data projects, including sequence-based predictive modeling for RNA modification prediction. Contributed to implementation and analysis supporting research on RNA 5-hydroxymethylcytosine (5hmC) prediction while continuing to develop practical data science and machine learning expertise.

Core Stack: Deep Learning, LSTM, Python, TensorFlow.

Joi Invent | Brisbane, Australia

Not-for-profit social enterprise and construction technology company with 250+ employees, pioneering 3D concrete printing in Queensland, delivering government-funded infrastructure projects with EY as the independent financial auditor.

Data Scientist | November 2018 - December 2019

- Contributed to the R&D team by researching and evaluating emerging data science and machine learning approaches to address construction and infrastructure challenges.

- Designed and prototyped end-to-end machine learning pipelines across six government-funded infrastructure projects, including data construction and characterization, feature engineering, model selection, and evaluation to deliver predictive insights and analytical frameworks.
- Collaborated with development and testing teams on solution design, integration, and deployment, supporting iterative build-test workflows and stakeholder reporting under government funding requirements.
- Translated model outputs and analytical findings into actionable insights for project managers, bridging computational methods and operational decision-making.
- Delivered product training to project managers and teams on the developed task management tool, supporting successful adoption and effective use across projects.

Core Stack: Analytics, Unstructured Data Analysis, Classification, Clustering, Recommendation, CNN, RNN, NLP, SQL, Python, Scikit-learn.

Griffith University | Gold Coast, Australia

Top-ranked global research university consistently ranked among the top 2% of universities worldwide, with doctoral research focused on machine learning, bioinformatics, and computational molecular science.

Ph.D. Candidate & Researcher | April 2014 - August 2018

- Constructed and curated protein complex datasets independently from protein sequence and structural data, designing data processing and feature-engineering workflows to support machine learning research.
- Developed and evaluated machine learning and deep learning models for protein and small biomolecular complex interaction and function prediction.
- Developed and deployed a publicly available ML-based protein-function prediction platform, integrating protein sequence and structural analysis with bioinformatics and machine learning methods.
- Presented part of research at ISMB (Intelligent Systems for Molecular Biology), the flagship international conference in bioinformatics and computational biology.

Core Stack: Classification, Clustering, Statistical testing, Protein sequence and structural analysis, Python, Keras, Scikit-learn, TensorFlow, Biopython, R, MATLAB, Bash, HTML, CSS, BLAST, PSI-BLAST.

EDUCATION

Ph.D., Computer Science - Applied Machine Learning & Bioinformatics

Griffith University - Gold Coast, Australia | 2014 - 2018

Dissertation: Protein Function Prediction by Machine Learning and Deep Learning.

M.Sc., Computer Science - Artificial Intelligence

University of Malaya - Kuala Lumpur, Malaysia | 2011 - 2013

Dissertation: Optimization of Neural Networks Using Evolutionary Algorithms.

B.Sc., Information Technology - Software Engineering

Multimedia University - Cyberjaya, Malaysia | 2007 - 2011

SELECTED PROJECTS & SCIENTIFIC IMPACT

40+ peer-reviewed publications | 1,407 citations | h-index 19

- Applied machine learning and molecular modeling to antibody-antigen and T-cell receptor docking and protein interactions, with research published in *PLOS Computational Biology*.
- Developed deep learning approaches for protein interaction and function prediction, with publications in *Applied Soft Computing*, *Bioinformatics*, and *Proteins: Structure, Function, and Bioinformatics*.
- Designed and developed an AI-driven medication adherence management platform integrating predictive modeling, explainability, causal analysis, and LLM/RAG-based decision support to support clinical outcomes, patient outreach, and healthcare decision-making at *PerformRx*.